

rename-plan

CONST-052 Rename Programme — HXC-001 Assessment + Phased Rename Plan

Revision: 1 **Last modified:** 2026-05-21T02:30:50Z **Description:** Current-state assessment of the CONST-052 lowercase-snake_case rename programme (HXC-001) plus a phased, atomic, low-risk→high-risk rename plan. **Authority:** Constitution §11.4.29 (CONST-052), §11.4.44 (this header); HXC-001 in docs/Issues.md. **Maintainer:** Operator + AI loop per §11.4.42 **Status:** Plan — research output, no renames executed.

0. Scope and method

CONST-052 (constitution §11.4.29) mandates every directory, submodule, and file under HelixCode use lowercase snake_case names, with technology- preserving common-sense exceptions. This document assesses the current state, distinguishes work already done (notably commit a1ea3c8 and its two sibling batch commits) from work remaining, applies the exemption carve-outs, maps the entanglement risk per rename candidate, and lays out a phased plan. **No renames were performed producing this document.**

A prior phased plan exists at docs/superpowers/specs/2026-05-19-const052-rename-programme-plan.md (commit f6664104, round 113). This document supersedes it as the current-state assessment: round-343 executed three of its batches and the inventory below reflects that progress.

Survey commands run 2026-05-21 against HEAD b4f2193 (4 remotes fetched, HEAD . .@{u} empty — local is current).

1. Current-state inventory

1.1 Already compliant (no action)

- **Meta-repo top-level directories** — all snake_case: helix_code/, helix_qa/, helix_agent/, assets/, challenges/, containers/, security/, constitution/, dependencies/, docs/, scripts/, cli_agents/, cli_agents_resources/, cli_agents_configs/, mcp_servers/, github_pages_website/, panoptic/, awesome-ai-memory/, implementation_guide/, upstreams/ (the meta-root one). Earlier rounds (15c54316 Dependencies/→dependencies/, cc339fc0 HelixCode/→helix_code/, 73cb32d2 HelixAgent/, 5ad402bb HelixQA/, 3f79f72a Containers/, af62a8cc Challenges/+Security/, a01c7c0e/17f84987 cmd/ + applications/) brought the top two layers into compliance.
- **Go module paths** — abstract(digital.vasic.auth, dev.helix.agent, etc.); they do NOT embed the directory name, so a directory rename does **not** change any module path. This materially lowers the risk class (see §3).

1.2 Already renamed by a1ea3c8 and sibling round-343 batches

Round 343 executed the safe (zero-go.mod-replace-consumer) leaf batches. All three confirmed present at new paths on disk and in `.gitmodules`:

Commit	Renamed	Count
a1ea3c8	submodules/models → models	1
416fe8e	submodules/ debate_orchestrator → debate_orchestrator	1
e813b5c	llvasic-digital/* leaves: auto_temp, claritas, doc_processor, gandalf_solutions, hyper_tune,i_llm, leak_hub,ouroborous, plinius_common, veritas, vision_engine	11

Total already done by round-343: 13 owned-org leaf directory renames. Each carried atomic `.gitmodules` path + section-name updates, `.git/modules/...` gitdir-pointer fixes, and ledger-row updates; `go build ./internal/... ./cmd/... exit 0` was captured per batch.

1.3 What remains — categorised

(A) Owned-org PascalCase submodule directories — 46 remaining.

`dependencies/HelixDevelopment/*` (8 PascalCase remain): DocProcessor, HelixLLM, HelixMemory, HelixSpecifier, LLMOrchestrator, LLMProvider, LLMsVerifier, VisionEngine.

`dependencies/vasic-digital/*` (38 PascalCase remain): Agentic, Auth, BackgroundTasks, Benchmark, Cache, Concurrency, Config, Database, Document, Embeddings, EventBus, Filesystem, Formatters, I18n, Lazy, LLMOps, LLMOrchestrator, LLMProvider, MCP_Module, Memory, Messaging, Middleware, Models, Normalize, Observability, Optimization, Planning, Plugins, RAG, RateLimiter, Recovery, RedTeam, SelfImprove, SkillRegistry, Storage, Streaming, ToolSchema, TOON, VectorDB, Watcher. (40 listed; MCP_Module and I18n are partially-snake/mixed — still non-compliant. Total non-compliant under `vasic-digital/` = 38 once MCP_Module→mcp_module and I18n→i18n and TOON→toon count.)

ALL 46 are **go.mod-entangled**: every one is referenced by a `replace <module> => ../dependencies/<Org>/<Dir>` directive in `helix_code/go.mod` and/or `helix_agent/go.mod` and/or `helix_qa/go.mod` (verified by `grep` of all three consumer `go.mod` files). This is the HIGH-RISK class — a rename without an atomic `replace-path` edit breaks the consumer build.

(B) Parent dependency directory `HelixDevelopment/`. Per round-343 operator decision D-2, the target is `helix_development/`. Renaming the parent dir changes the `../dependencies/HelixDevelopment/...` path component in EVERY consumer replace directive that points into it (12 directives across `helix_code/go.mod` + `helix_agent/go.mod`) plus every `.gitmodules` path = line for a `HelixDevelopment/*` submodule (10 entries). `vasic-digital/` is KEPT (D-3: GitHub-handle / proper-noun carve-out — it matches the literal org name).

(C) Upstreams/ → upstreams/ — 59 directories. One inside the meta-repo’s own submodule trees and inside owned-org submodule trees. Each `Upstreams/` lives INSIDE a submodule’s git tree, so each rename is a commit in that submodule’s repo, then a `.gitmodules` pointer bump in the parent. The meta-root `upstreams/` is already lowercase. Note: `constitution/install_upstreams.sh` already supports BOTH `Upstreams/` and `upstreams/` (constitution commit 45d3678), so this cluster does NOT break the upstream-install tooling mid-flight.

(D) .gitmodules entries. Path/name segments tracking (A)/(B)/(C): 46 owned-org submodule sections still carry PascalCase path = and [submodule "..."] name segments. Third-party submodule sections (`cli_agents/*`, `cli_agents_resources/*`, `dependencies/LLama_CPP`, `dependencies/Ollama`, `dependencies/HuggingFace_Hub`, `awesome-ai-memory`) are EXEMPT (see §2).

(E) Files. UPPERCASE tracked `.md` files are widespread (`COMPREHENSIVE_*.md`, `PHASE_*.md`, `FINAL_*.md`, etc. — ~120 at meta-root, dozens under `helix_code/`). Standard-metadata files (`README.md`, `CLAUDE.md`, `AGENTS.md`, `CONSTITUTION.md`, `LICENSE`, `CONTINUATION.md`) are EXEMPT per §11.4.44 OUT-scope + de-facto convention. The remaining UPPERCASE_SNAKE report/plan `.md` files at meta-root and under `helix_code/` are technically in CONST-052 scope for files but are **deferred to a final low-risk cosmetic phase** (no build/import entanglement; reference cost is internal cross-links only) — see Phase 6.

2. Exemption list (stays as-is — technology-preserving carve-outs)

Per CONST-052 §11.4.29 the “does renaming break the technology?” test trumps the rule. EXEMPT:

Item	Why exempt
<code>cli_agents/*</code> (45+ submodules: <code>aider</code> , <code>crush</code> , <code>claude-code</code> , ...)	Third-party vendor submodules — keep upstream repo names. Mostly already lowercase anyway.
<code>cli_agents_resources/*</code> (<code>Awesome-AI-Agents</code> , <code>Awesome-AI-GPTs</code> , <code>Cheshire-Cat-Ai</code> , <code>GitHub-Awesome-Copilot</code> , <code>OpenAI-Cookbook</code> , <code>Taches-CC-Resources</code>)	Third-party vendor submodules (<code>github/</code> , <code>openai/</code> , <code>e2b-dev/</code> , ...) — keep upstream names.
<code>dependencies/LLama_CPP</code> , <code>dependencies/Ollama</code> , <code>dependencies/HuggingFace_Hub</code>	Third-party vendor submodules (<code>ggml-org/</code> , <code>ollama/</code> , <code>huggingface/</code>).

Item	Why exempt
awesome-ai-memory/	Third-party (topoteretes/). Already lowercase.
vasic-digital/ parent dir	Literal GitHub-org handle — proper-noun carve-out (operator decision D-3, round 343).
README.md, LICENSE, CLAUDE.md, AGENTS.md, CONSTITUTION.md, CRUSH.md, QWEN.md, CONTINUATION.md, NOTICE, VERSION, OWNERS	Standard-metadata / governance files — conventional uppercase; §11.4.44 lists CLAUDE/AGENTS/README as OUT-scope; renaming breaks tool discovery (Claude Code, golangci, etc.).
Language-mandated subtrees	Java/Kotlin/Android/Apple/Swift/C# source roots inside helix_code/applications/{android,ios,aurora-os,harmony-os} — file/dir case follows the language convention; submodule root still follows our convention.
Build artefacts	node_modules/, .git/, target/, build/, bin/, __pycache__/ — tool-mandated names.
Go package directories already lowercase	Go requires lowercase import-path segments anyway; internal/ packages are compliant.

Note on third-party Upstreams/: the 59-count in §1.3(C) includes Upstreams/ dirs inside third-party submodule trees only where those trees are owned-by-us (the count was taken across owned submodules). cli_agents/claude-code-source/Upstreams — claude-code-source is the operator's own private mirror (gitlab.com:milos85vasic/ccode-private) so its Upstreams/ IS in scope. Pure third-party Upstreams/ (if any) are exempt.

3. Risk map

Rename class	Entanglement	Risk
Owned-org leaf dir (A)	.gitmodules path + section name; .git/modules/... gitdir pointer; consumer go.mod replace path; coverage ledgers; governance-doc prose	HIGH — replace-path edit must be atomic with the rename or the consumer build breaks. NOT a module-path change (paths are abstract), so go list -m output is unchanged; the break is purely replace filesystem-path resolution.
Parent dir HelixDevelopment/ →helix_development/ (B)	ALL 12 replace directives pointing into it + ALL 10 HelixDevelopment/* .gitmodules path=	HIGHEST — single rename, broad atomic fan-out across 2 consumer go.mod files + 10

Rename class	Entanglement	Risk
	lines + <code>.git/modules/dependencies/HelixDevelopment/ tree</code>	submodule sections. Must be one PWU.
Upstreams/ →upstreams/ (C)	Inside each submodule's own git tree; needs an inner-repo commit then a parent <code>.gitmodules</code> pointer bump	MEDIUM — 59× (inner commit + parent bump). Low risk per commit, high commit count. No <code>go.mod</code> entanglement. Tooling already dual-name-safe.
<code>.gitmodules</code> segment (D)	Folds into (A)/(B) — same commit	covered by A/B
UPPERCASE <code>.md</code> files (E)	Internal markdown cross-links only; no build/import impact	LOW — cosmetic; deferred to Phase 6.

High-risk count: 47 = 46 owned-org leaf dirs (A) + 1 parent dir (B). Every one touches a Go `replace` path and therefore needs a build-verified atomic commit. None changes a Go *module path*, so none requires editing `import` statements in `.go` source — that is the one piece of good news that keeps this from being catastrophic.

4. Phased rename plan (low-risk → high-risk)

Each phase = a set of fine-grained atomic rename tasks. Each task lists the rename, every reference updated atomically, and the test verification (§11.4.29 mandates each batch ships a regression test verifying every reference resolves + a full test-type run).

Phase 0 — Pre-flight tooling (no renames)

- **Task 0.1** — Author `scripts/const052_rename_leaf.sh <org> <Old> <new>`: performs `git mv`, rewrites `.gitmodules` (path + section name), fixes `.git/modules/.../ gitdir` + `worktree` pointer, rewrites the matching `replace` line(s) in every consumer `go.mod`, updates `docs/coverage/COVERAGE_LEDGER.md` + `docs/coverage/ledger.md` rows. Idempotent dry-run flag.
- **Task 0.2** — Author `scripts/const052_verify_refs.sh`: the regression test — greps the whole tree for the OLD path string (must be 0 hits outside historical CONTINUATION prose), runs `git submodule status (all resolve)`, runs `cd helix_code && go build ./... && cd ../helix_agent && go build ./... (exit 0)`.
- **Verification:** dry-run the tooling against an already-renamed leaf (`models`) — must report “no change needed”. Paired-mutation: plant a stale ref, assert the verifier FAILs.

Phase 1 — Upstreams/→upstreams/ cluster (C), 59 dirs — MEDIUM, no go.mod

Independent of the go.mod work; can run first as a low-risk warm-up. Group into batches of ~10 submodules.

- **Task 1.x (per submodule)** — inside <submodule>/: `git mv Upstreams upstreams`, commit + push the submodule's 4 remotes, then in the parent bump the `.gitmodules` pointer for that submodule in the SAME parent commit. Update any `install_upstreams` invocation docs that hardcode `Upstreams/` (tooling itself is already dual-name-safe).
- **References per task:** submodule-internal git tree only + parent `.gitmodules` SHA pointer. No `replace`, no `import`, no `config`.
- **Verification:** `install_upstreams` from the renamed submodule root resolves all recipes; `git remote -v | grep -c push` reports expected count; submodule status clean. Batch ships the §11.4.29 regression test.

Phase 2 — owned-org leaf dirs with EXACTLY ONE consumer (A subset) — HIGH but minimal fan-out

The §1.3(A) `replace-count` showed most `vasic-digital/*` leaves have a single consumer `replace` directive. Batch ~8–10 per round.

- **Task 2.x (per dir)** — e.g. `vasic-digital/Cache→cache`: `git mv submodules/cache submodules/cache`; rewrite the ONE `replace digital.vasic.cache => ../submodules/cache` line in `helix_agent/go.mod`; rewrite `.gitmodules` path + section name; fix `.git/modules/...` pointer; update coverage ledgers; sweep governance docs (CLAUDE.md §3.2 prose, docs/ references) — active references only, historical CONTINUATION prose left intact.
- **Verification per batch:** `const052_verify_refs.sh` (0 stale refs, submodule status resolves, `go build ./... exit 0` for every consumer); full test-type run on affected consumers (`make test` in `helix_agent`).

Phase 3 — owned-org leaf dirs with MULTIPLE consumers (A subset) — HIGH

Leaves referenced by 2–3 consumer `go.mod` files: `VisionEngine`, `LLMOrchestrator`, `DocProcessor`, `Memory`, `LLMsVerifier`, `HelixSpecifier`. Each rename must rewrite the `replace` line in EVERY consumer atomically.

- **Task 3.x (per dir)** — same as Phase 2 but the `replace`-rewrite spans all consumers (e.g. `DocProcessor` appears in both `helix_code/go.mod` and `helix_agent/go.mod`). One PWU per dir.
- **Verification:** build ALL consumers; full test-type run on each.

Phase 4 — MCP_Module, I18n, TOON, LLM0ps, RAG normalisation — HIGH

Mixed/acronym dirs needing operator-blessed `snake_case` targets (D-6 `mcp_module`, D-7/D-8/D-9 already decided as `i18n/toon/rag` style). Same atomic procedure as Phase 2/3.

Phase 5 — parent dir HelixDevelopment/→helix_development/ (B) — HIGHEST

Single rename, broadest atomic fan-out. ONE PWU.

- **Task 5.1** — `git mv dependencies/HelixDevelopment dependencies/helix_development`; rewrite ALL 12 `replace ../dependencies/HelixDevelopment/...` lines (across `helix_code/go.mod + helix_agent/go.mod`) to `../helix_development/...`; rewrite ALL 10 `HelixDevelopment/*.gitmodules path= + section-name` segments; move the `.git/modules/dependencies/HelixDevelopment/` subtree; sweep all governance docs + ledgers.
- **Verification:** every consumer `go build ./...` exit 0; all 10 submodule statuses resolve; full test-type matrix run; §11.4.29 regression test.

Phase 6 — UPPERCASE .md file renames (E) — LOW, cosmetic

Rename non-exempt UPPERCASE report/plan .md files at meta-root and under `helix_code/` to `snake_case`. Batch by directory.

- **Task 6.x** — `git mv COMPREHENSIVE_ANALYSIS_REPORT.md comprehensive_analysis_report.md`; `grep + rewrite` every internal markdown cross-link. NO build impact.
- **Verification:** link-checker confirms no broken relative links.

5. Test coverage required per batch (CONST-052 §11.4.29 + CONST-050(B))

Every rename batch ships:

1. **Regression test** — `scripts/const052_verify_refs.sh`: 0 stale OLD-path references outside historical prose; all `git submodule status` resolve; all consumer `go build ./...` exit 0.
2. **Full test-type run** — `make test` (unit) on every affected consumer module; integration/E2E where the renamed submodule is exercised; anti-bluff captured runtime evidence (paste exit codes + output).
3. **Paired mutation** — plant a known stale reference, assert the regression test reports FAIL (proves the test is not a bluff).

6. Parallel-session coordination risk (honest note)

Round-343 batch-1/2/3 (a1ea3c8, 416fe8e, e813b5c) were executed by a prior session. Two concrete risks for the next executor:

- **Stale local state.** Per CONST-060 the first `git` action MUST be `git fetch --all --prune + git log HEAD..@{u} + recursive submodule fetch`. Acting on a stale tree risks re-renaming an already-renamed leaf or colliding on `.gitmodules`. (At the time of writing, `HEAD..@{u}` is empty — local is current.)
- **.gitmodules is a single-file collision hotspot.** Every phase edits `.gitmodules`; two parallel sessions WILL conflict there. Recommendation: serialise the rename programme to

ONE session, or partition strictly by phase with an explicit hand-off, and always fetch-first per §11.4.71/CONST-060 before each batch.

- **Submodule-internal commits (Phase 1 cluster C)** land in separate repos with their own 4-remote push; a parent `.gitmodules` pointer bump **MUST NOT** be pushed until the submodule push is accepted by all remotes (per the prior plan's §7.4).
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7. Summary

- **Rename candidates remaining:** 46 owned-org leaf dirs (A) + 1 parent dir (B) + 59 `Upstreams/` dirs (C) + N UPPERCASE `.md` files (E, deferred cosmetic). Core structural total = **106** dir-level renames.
- **Already done by a1ea3c8 + siblings:** 13 owned-org leaf renames.
- **Phases:** 7 (Phase 0 pre-flight through Phase 6 cosmetic).
- **High-risk (go.mod-replace-entangled):** 47 (46 leaves + 1 parent).
- **Good news:** Go module paths are abstract — no `.go import` statement edits required; the entanglement is `replace` filesystem paths only.